GU-GS: Gaussian Splatting-Based Geometry Refinement and Uncertainty-Aware Learning Method for DSM Generation From Satellite Imagery

Bin Ding, Junyuan Liu, Xiaoran Yin, Zi Yang, Yunfeng Luo, and Daijun Zhang

***Abstract*—The 3-D reconstruction from multitemporal satellite imagery is a critical task. While 3-D Gaussian splatting (3DGS) has emerged as an efficient alternative to neural radiance field (NeRF)-based methods, its direct application to satellite data often produces geometric noise and artifacts (e.g., floaters and blurry surfaces) and is highly sensitive to illumination and shadow changes, hindering high-fidelity digital surface model (DSM) extraction. To address these limitations, we introduce GU-GS, a Gaussian splatting-based geometry refinement (GR) and uncertainty-aware learning (UAL) method for accurate and efficient DSM generation from satellite imagery. GR module employs a significance-guided pruning strategy to remove redundant Gaussian primitives and leverages opacity entropy regularization to eliminate blurry and unstable surfaces. Through this process, we construct a compact and geometrically consistent primitive set, thus reducing the geometric noise. UAL module leverages a predictor based on semantic features to identify unreliable regions caused by multitemporal illumination and shadow changes and generates an uncertainty mask. By dynamically down-weighting these regions' contribution to the photometric loss, it effectively filters the supervision signal during the training process, thereby enhancing the stability of optimization and the accuracy of geometric reconstruction. Experiments on the DFC2019 and IARPA2016 datasets demonstrate that our method generates high-precision DSMs, exhibiting superior performance in both accuracy and training speed compared to existing approaches. Compared with Sat-NeRF, our method achieves an average accuracy improvement of 11% and a training speedup of 40×; relative to EOGS, it shows an average accuracy enhancement of 24% and a training speed improvement of 1.8×. Code will be available at: https://github.com/thessshy/GU-GS**

***Index Terms*—3-D Gaussian splatting (3DGS), digital surface model (DSM), photogrammetry, remote sensing, satellite image.**

Received 30 September 2025; revised 15 January 2026; accepted 3 March 2026. Date of publication 6 March 2026; date of current version 17 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62173320 and in part by the Institute's Key Project on Basic Reinforcement under Grant E3290401. *(Corresponding authors: Daijun Zhang; Junyuan Liu.)*

Bin Ding and Junyuan Liu are with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang 110016, China, and also with the School of Computer Science and Technology, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: dingbin@sia.cn; liujunyuan@sia.cn).

Xiaoran Yin, Zi Yang, Yunfeng Luo, and Daijun Zhang are with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang 110016, China (e-mail: yinxiaoran@sia.cn; yangzi@sia.cn; luoyunfeng@sia.cn; zhangdaijun@sia.cn).

Digital Object Identifier 10.1109/TGRS.2026.3671346

I. Introduction

HIGH-RESOLUTION satellite imagery plays a pivotal role in various remote sensing applications, including urban infrastructure monitoring, environmental analysis, and disaster response [1], [2], [3], [4]. One of the essential tasks enabled by such data is the reconstruction of the Earth's surface geometry in the form of digital surface models (DSMs) [5], which encode both terrain and artificial structures. Traditional DSM generation methods typically rely on stereo or tristereo satellite image pairs [6], [7]. These approaches utilize photogrammetric pipelines and disparity estimation algorithms such as semiglobal matching (SGM) to recover depth. However, their effectiveness depends heavily on near-simultaneous image acquisition and favorable imaging geometry, which are not always available in practical scenarios.

As satellite constellations continue to expand, a growing volume of multiview and multidate satellite imagery has become available. Despite their rich spatial and temporal content, such datasets remain underexploited. Many existing methods [6] still treat image pairs independently, failing to fully exploit the consistency across views. Moreover, dense stereomatching methods [8] based on hand-crafted features are prone to errors in textureless areas, regions with occlusion, and under varying illumination conditions [9].

Learning-based approaches have demonstrated improved robustness by learning discriminative features from data [10]. However, most supervised models require large amounts of ground-truth DSMs, such as those generated by LiDAR. This dependency limits their scalability and generalization to new geographical regions or satellite sensors, thus impeding large-scale deployment. In recent years, neural radiance fields (NeRFs) [11] have emerged as a novel paradigm for 3-D scene reconstruction, offering photorealistic rendering and detailed geometry estimation through learned volumetric representations. Adaptations of NeRF to Earth's observation, such as Sat-NeRF [12], have shown promising results for satellite-based surface modeling. However, NeRF-based models are computationally intensive and require long training times, which becomes a significant barrier for high-resolution, large-area satellite data [13], [14], [15].

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

To address these limitations, 3-D Gaussian splatting (3DGS) [16] has recently been introduced as a compelling alternative. It represents the scene as a collection of anisotropic Gaussians in 3-D space, supporting efficient training and real-time rendering. Compared to NeRF, 3DGS achieves the equivalent reconstruction accuracy while significantly reducing the computational overhead, making it more suitable for operational deployment on large-scale datasets [17], [18], [19]. However, a fundamental incompatibility arises when directly applying vanilla 3DGS, which is inherently designed for static scenes, to multitemporal satellite imagery. Multitemporal satellite observations inherently capture dynamic scene changes, varying atmospheric conditions, transient objects, and inconsistent radiometry across different acquisition times. This fundamental mismatch leads to significant challenges for standard 3DGS, resulting in prevalent geometric artifacts such as floating Gaussians, ambiguous surfaces, and noisy DSMs that fail to represent the stable underlying scene geometry. These imperfections severely limit its utility for high-fidelity 3-D mapping and downstream applications.

In this article, we introduce GU-GS, a novel Gaussian splatting-based geometry refinement and uncertainty-aware learning (UAL) method for accurate and efficient DSM generation from satellite imagery, which significantly enhances the reconstruction quality and robustness in satellite photogrammetry. Specifically, it refines the 3-D representation by mitigating internal artifacts and enhances training stability by robustly handling external data corruption caused by photometric inconsistencies. As demonstrated in our experiments on the DFC2019 [20], [21], and IARPA2016 [22] datasets, GU-GS achieves state-of-the-art (SOTA) geometric accuracy, outperforming traditional, NeRF-based, and previous 3DGS-based methods in DSM generation. Our main contributions are summarized as follows.

1) We propose a two-stage geometry refinement module to improve model compactness and geometric fidelity. This module first performs a coarse pruning of structurally unimportant primitives, effectively eliminating redundant Gaussians that contribute to noise and excessive memory usage. It then applies a continuous opacity entropy regularization to resolve ambiguities and refine surface geometry, addressing the problem of boundary artifacts prevalent in vanilla 3DGS reconstructions.
2) We introduce a UAL module that leverages robust semantic features from the DINOv2 to identify unreliable regions caused by multitemporal illumination and shadow changes, thereby masking their influence on the supervision signal without destabilizing the main optimization. By mitigating the detrimental impact of these inconsistent data during the training process, our module significantly improves the accuracy of the reconstructed scene.
3) We demonstrate that the integration of these specialized modules jointly addresses the unique challenges of satellite imagery, leading to superior geometric accuracy and computational efficiency in DSM generation.

This article is structured as follows. Section II reviews the relevant prior work in satellite 3-D reconstruction and neural rendering. Section III then elaborates on our proposed GU-GS methodology. A series of experiments and analyses, including ablation studies, is presented in Section IV to demonstrate the effectiveness and advantages of our method. Finally, Section V concludes this article.

II. Related Works

In this section, we review stereovision satellite 3-D reconstruction, NeRF variants for multiview satellite imagery, and recent advancements in 3DGS.

A. Stereovision for Satellite Imagery

Stereovision is a fundamental approach for generating 3-D surface models from satellite imagery and serves as the core for several well-established photogrammetric pipelines, including S2P [6] and CARS [7]. These systems typically follow a sequential process involving image pair selection, epipolar rectification based on rational polynomial coefficient (RPC) models, dense disparity computation, triangulation, and depth map fusion. Among the disparity estimation techniques, SGM [8] and its variants, such as more global matching (MGM) [23], semiglobal block matching (SGBM) [24], and tSGM [25], have been widely adopted due to their balance between computational efficiency and reconstruction accuracy, particularly in areas with weak texture or depth discontinuities.

To overcome the limitations of traditional artificial matching costs and improve robustness against variations in illumination and seasonal conditions, recent research has increasingly turned to deep learning-based stereo methods [10], [26], [27]. For example, the integration of GA-Net [28] into the S2P framework has demonstrated improved disparity estimation under radiometric noise [29]. Other methods, including dual-scale networks [30], DSMNet [31], and HMSM-Net [32] employ encoder–decoder architectures and hierarchical or multiscale cost volume designs to improve the performance in low-texture or occluded regions. Semantic-guided models such as BGA-Net [33] further enhance reconstruction reliability by incorporating semantic cues and multitask learning to adapt to appearance changes across time and season. However, most of these approaches are fundamentally pairwise and do not fully exploit the redundancy inherent in multiview satellite datasets. To overcome this limitation, recent studies have explored the adaptation of multiview stereo (MVS) techniques for satellite image processing [34], [35], [36]. Some methods approximate the complex imaging geometry of satellites with pinhole camera models to enable the use of conventional MVS frameworks such as COLMAP [37], [38]. Other approaches have introduced deep learning-based MVS pipelines tailored to the satellite domain, such as Sat-MVSF [39], which combines preprocessing, volumetric reconstruction, and refinement in a unified architecture.

Despite these advancements, conventional stereovision methods, relying heavily on pixel-level matching, remain sensitive to inconsistent parallax and radiometric variation inherent in multitemporal satellite imagery. These limitations, arising from wide baselines, nonsimultaneous acquisitions, and significant seasonal or illumination differences, highlight the need for more robust reconstruction methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

B. NeRF for Satellite Imagery

NeRF [11] has recently emerged as a powerful framework for novel view synthesis and 3-D scene reconstruction by implicitly modeling volumetric density and color through a multilayer perceptron (MLP) [40]. While originally developed for static, ground-based scenes, a growing body of research has adapted NeRF for satellite imagery, which introduces new challenges such as wide baselines, multidate observations, lighting variations, and limited viewing angles.

S-NeRF [14] augments the original NeRF framework by explicitly incorporating solar illumination parameters, such as solar azimuth and elevation angles, into the volumetric radiance field representation. This allows the model to account for directional lighting effects and accurately model cast shadows, thereby improving the fidelity of illumination-aware 3-D reconstructions. Building upon this, Sat-NeRF [12] introduces a more realistic imaging geometry by leveraging RPC sensor models, which better reflect the physical characteristics of satellite image acquisition. In addition, it incorporates a transient-object modeling branch that estimates per-ray uncertainty to attenuate the influence of dynamic scene elements, such as vehicles or foliage [41], [42]. EO-NeRF [15] further refines the geometric consistency by simulating shadow generation through physically based ray tracing from reconstructed surfaces toward the solar direction. This strategy enables the generation of geometrically coherent shadows and leads to significant improvements in DSM accuracy compared with both traditional stereo pipelines and preceding NeRF-based methods.

Recent developments have increasingly focused on improving the computational efficiency and generalization capabilities of NeRF-based frameworks for satellite imagery. RS-NeRF [43] introduces a lightweight architecture by employing hash-based positional encoding and compact network design, substantially accelerating both training and inference. Sat-Mesh [44] incorporates photometric consistency constraints across diverse solar angles, using them as supervisory signals to guide accurate surface reconstruction. Sat-NGP [45] adopts a hash-grid representation inspired by instant neural graphics primitives to achieve rapid convergence. FVMD-ISRe [46] generates complete watertight meshes and DSMs from few-view multidate satellite images by using an adaptive frequency strategy to handle insufficient geometric constraints and incorporating solar data to resolve illumination differences. Sat-DN [47] leverages depth and normal supervision within a multiresolution hash-grid framework to enhance geometric accuracy under illumination variations and weak textures, achieving more consistent planar surfaces and improved reconstruction quality from multiview satellite imagery.

C. 3-D Gaussian Splatting

Recently, 3DGS [16] has introduced significant advances in neural rendering [48], [49]. As a highly efficient alternative to NeRF, 3DGS explicitly represents a scene using a dense set of optimizable 3-D Gaussian primitives $\{\mathcal{G}_i\}$. Novel views are synthesized via a differentiable rasterizer that projects these 3-D Gaussians onto the 2-D image plane, followed by an alpha-blended accumulation. This explicit representation significantly accelerates both training and rendering.

1) 3DGS for Satellite Imagery: While 3DGS has demonstrated impressive real-time novel view synthesis for general scenes, recent studies have explored its adaptation to the unique challenges of Earth observation imagery. EOGS [18] is the first approach to apply 3DGS for digital elevation modeling. It adapts 3DGS to remote sensing tasks by integrating radiometric correction and shadow modeling, achieving accuracy comparable with SOTA methods with significantly reduced computational costs. Although also built upon the 3DGS framework, Sat-GS [50] focuses on the novel view synthesis for multitemporal satellite imagery, addressing appearance variations due to seasonal changes and transient objects. SkySplat [51] further extends 3DGS to multitemporal sparse-view satellite imagery by introducing a generalizable RPC-based reconstruction framework that suppresses transient objects and leverages height supervision, enabling fast and accurate DSM generation across diverse scenes.

2) Pruning and Compression of Gaussian Primitives: Despite its efficiency, the densification strategy in 3DGS often results in a massive number of Gaussian primitives, leading to significant storage costs and memory overhead. This redundancy can also introduce geometric noise, such as floaters and ambiguous surfaces, which is particularly detrimental for high-precision DSM extraction. Consequently, several studies have focused on pruning or compressing these representations. Beyond the simple opacity-based culling in the original method, more advanced criteria have been proposed. Reduced3DGS [52] proposes a resolution-aware pruning strategy. It identifies regions where the primitive density exceeds the detail level resolvable by the training cameras and culls redundant Gaussians based on intersection tests within these regions. TrimGS [53] and LightGaussian [54] prune 3-D Gaussians by calculating an individual contribution score for each primitive based on its role in the alpha-blending rendering process. In contrast, a more global approach SUNDAE [55] models the spatial relationships between all primitives by constructing a graph and then uses spectral analysis to collectively downsample the set of Gaussians. A key distinction in these approaches is their ultimate goal: some prioritize model compression for efficient storage, while others, like TrimGS [53], focus on the geometric accuracy by controlling primitive scales. However, these general-purpose techniques are not specifically tailored to the unique geometric artifacts present in satellite photogrammetry. Our work is motivated by the need for a refinement strategy that addresses the unique geometric artifacts and noise in the 3-D representation for high-fidelity DSM generation.

III. Method

In this section, we introduce a two-pronged strategy to address geometric artifacts and photometric inconsistencies arising from multitemporal variations, as illustrated in Fig. 1. The first component is a geometry refinement module that refines the 3-D representation through significance-guided

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

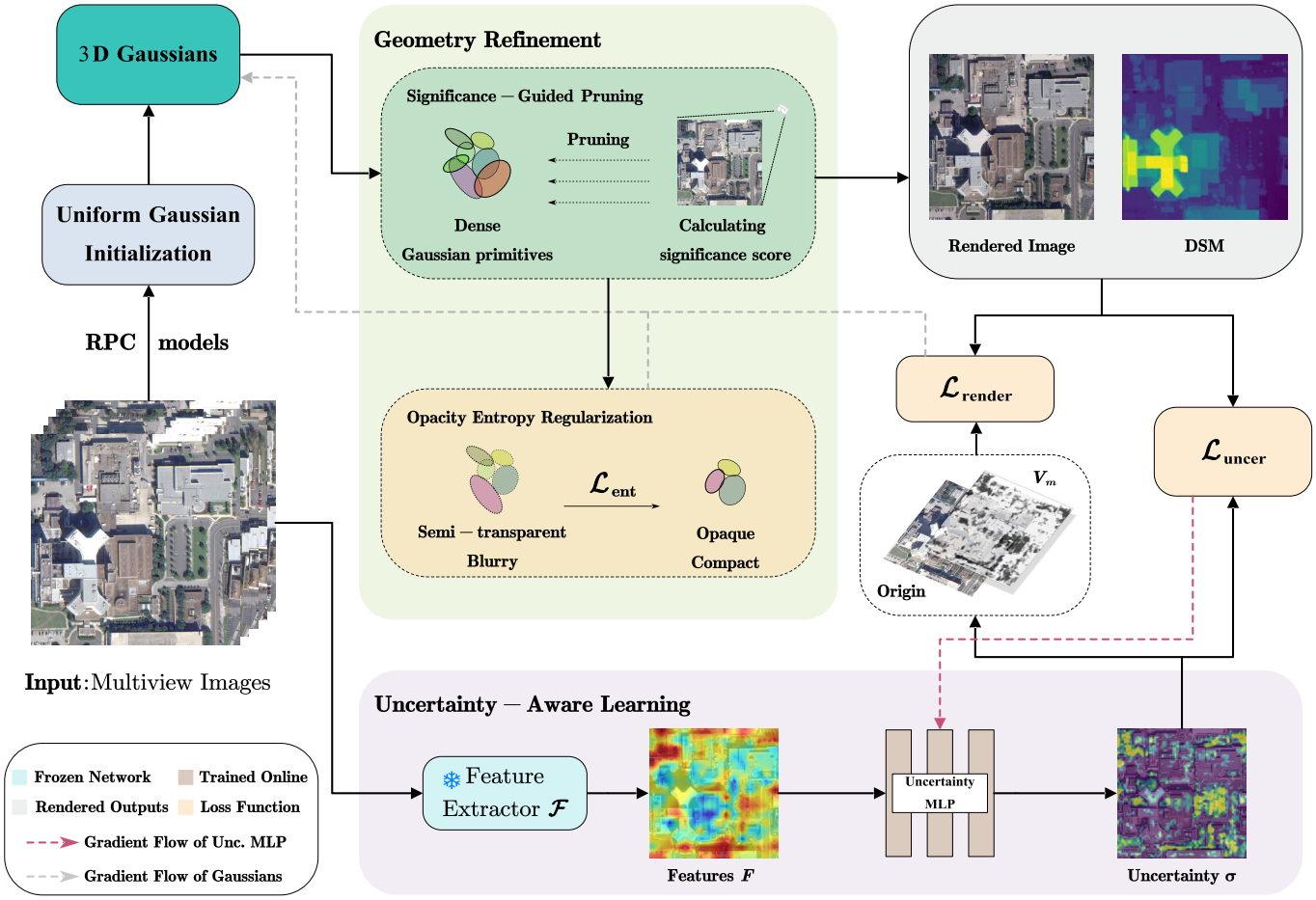


Fig. 1. Overview of our method. Our method takes multiview satellite images as input to reconstruct a high-fidelity 3-D scene represented by 3-D Gaussians. Input multiview images are processed using RPC models to define the scene bounds. Subsequently, a uniform volumetric initialization of 3-D Gaussians is applied to ensure the complete coverage [18]. To address geometric artifacts and view-inconsistent data common in satellite imagery, we introduce a two-pronged strategy. The GR module operates on the 3-D Gaussians directly, employing a significance-guided pruning step to remove redundant primitives, followed by an opacity entropy regularization $\mathcal{L}_{ent}$ to resolve surface ambiguities and create a refined, geometrically consistent representation. In parallel, the UAL module leverages a pretrained feature extractor and a lightweight, online-trained MLP to predict a per-pixel uncertainty map. This map identifies unreliable regions, which are then down-weighted via the visibility map V_m in the rendering loss $\mathcal{L}_{render}$. Notably, the training of the uncertainty MLP and the 3-D Gaussians is decoupled; as indicated by different gradient flows, the uncertainty loss $\mathcal{L}_{uncer}$ only updates the MLP, ensuring training stability.

pruning and opacity entropy regularization (as detailed in Section III-B). The second is a UAL module, which leverages a feature-based predictor to identify and mitigate the influence of unreliable data during optimization (as detailed in Section III-C).

A. Preliminaries

The scene representation in 3DGS [16] consists of K independent Gaussian primitives, each associated with a corresponding 3-D Gaussian kernel $\mathcal{G}_k$

$$\mathcal{G}_k(\boldsymbol{x}) = e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)} \tag{1}$$

where $\boldsymbol{\mu}_k \in \mathbb{R}^3$ is the position center and $\boldsymbol{\Sigma}_k \in \mathbb{R}^{3\times3}$ is the covariance matrix. During rendering, these 3-D primitives are projected onto the 2-D image plane. The color C for a pixel u is then rendered by alpha-blending the primitives that overlap with it, sorted by depth

$$C(u) = \sum_{k\in N} T_k \alpha_k c_k, \quad T_k = \prod_{j=1}^{k-1}(1-\alpha_k) \tag{2}$$

where $\boldsymbol{c}_k \in \mathbb{R}^3$ is the color and $\alpha_k \in [0,1]$ its opacity.

To approximate the complex pushbroom imaging geometry of satellite sensors, EOGS replaces the full-RPC model with a per-scene affine transformation $\pi : x \in \mathbb{R}^3 \to Ax + a \in \mathbb{R}^2$. Each 3-D Gaussian primitive is projected from world coordinates into 2-D normalized device coordinates (NDC) space as follows:

$$\boldsymbol{\mu}_k^\pi = A\boldsymbol{\mu}_k + \boldsymbol{a}, \quad \boldsymbol{\Sigma}_k^\pi = A\boldsymbol{\Sigma}_k A^\top \tag{3}$$

where $A \in \mathbb{R}^{2\times3}$ and $\boldsymbol{a} \in \mathbb{R}^2$ are the affine parameters estimated per scene. This approximation simplifies the entire projection pipeline and eliminates the need for per-point Jacobian linearization while maintaining subpixel accuracy. For the task of DSM generation, each primitive's altitude is represented as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

$h_k \in \mathbb{R}$. Adapting the 3DGS rendering paradigm (2) to output elevation, the elevation rendering is defined as follows:

$$H(u) = \sum_{k \in N} T_k \alpha_k h_k. \tag{4}$$

The elevation rendering utilizes alpha-blending accumulation that terminates when transmittance drops below a threshold. The final DSM is generated by unprojecting the altitude map to a 3-D point cloud and rasterizing it onto a grid matching the input ground sampling distance, using weighted average pooling with a 1-pixel radius.

The framework is trained by minimizing a photometric reconstruction loss across multiview images

$$\mathcal{L}_{rgb} = (1-\lambda)\mathcal{L}_1\left(\tilde{C}, C_i\right) + \lambda \mathcal{L}_{D-SSIM}\left(\tilde{C}, C_i\right) \tag{5}$$

where $\tilde{C}$ is the rendered color, C_i is the ground-truth color, and $\lambda \in [0, 1]$ balances pixelwise and perceptual structural similarity terms. The $\mathcal{L}_1$ term enforces accurate color reconstruction, while $\mathcal{L}_{D-SSIM}$ promotes the structural consistency.

Furthermore, for enhancing the geometric stability and reducing redundancy, we apply an auxiliary regularization loss $\mathcal{L}_{reg}$, following the strategy proposed in EOGS. This loss consists of terms for opacity sparsity, local view consistency, and shadow opaqueness

$$\mathcal{L}_{reg} = \lambda_s \mathcal{L}_{sparsity} + \lambda_v \mathcal{L}_{view} + \lambda_o \mathcal{L}_{opaque} \tag{6}$$

where $\mathcal{L}_{sparsity}$ imposes an L1 penalty on Gaussian opacities to encourage a sparse distribution, $\mathcal{L}_{view}$ promotes the local view consistency by minimizing the color and elevation discrepancies between input views and randomly perturbed virtual views, and $\mathcal{L}_{opaque}$ mitigates semitransparent shadows by enforcing binary shadow factors. For the detailed mathematical derivation of these terms, we refer the reader to [18].

B. Geometry Refinement of Gaussian Primitives

EOGS adopts a simplified training pipeline by forgoing densification and relying solely on opacity-based pruning. Specifically, a large set of Gaussians is uniformly initialized at the beginning of optimization, and primitives with opacity below a fixed threshold are automatically discarded. This strategy, while effective at removing completely transparent Gaussians, leaves a considerable number of primitives with intermediate opacities that persist throughout training.

These partially transparent Gaussians often correspond to low-contribution regions, such as empty space, transient occluders, or view-inconsistent features. Although they may have a negligible impact on photometric rendering, they introduce significant noise when extracting explicit surface representations, such as DSM. In particular, such primitives tend to produce floating artifacts and undesired depth estimates in regions where no stable geometry exists. Moreover, in large-scale remote sensing scenarios, the presence of redundant Gaussians imposes additional memory and computational overhead. Since 3DGS uses an explicit representation where each primitive is individually stored and optimized, the total cost grows approximately linearly with the number of Gaussians. For satellite image collections covering urban areas or large-scale scenes, this leads to increased training time and resource consumption, and hinders scalability to higher resolution inputs. To address this fundamental limitation, we introduce a two-stage geometry refinement module.

1) Significance-Guided Pruning: Building upon previous works [53], [54], we introduce a significance-guided approach that goes beyond simple opacity thresholding. Rather than relying solely on opacity α, we formulate a significance score that holistically considers three key factors: multiview geometric consistency, rendering contribution, and spatial scale diversity. We assess each Gaussian's contribution to the rendered geometry based on its spatial support and accumulated visibility across views. By explicitly quantifying per-primitive importance, we remove redundant or unstable Gaussians, yielding a more compact and geometrically consistent representation suitable for downstream DSM generation.

To evaluate each Gaussian's contribution, we first identify the set of pixels it influences across all training views. Let $\mathcal{P}_k$ be the set of all pixel-view pairs (p, v) where the 2-D projection of Gaussian $\mathcal{G}_k$ covers pixel p in view v. This set is determined efficiently during the forward rasterization process. We then define the significance score $\mathrm{GS_k}$ for Gaussian $\mathcal{G}_k$ as follows:

$$\mathrm{GS}_k = |\mathcal{P}_k| \cdot \alpha_k \cdot \gamma(\mathbf{\Sigma}_k) \tag{7}$$

where $|\mathcal{P}_k|$ is the cardinality of the set $\mathcal{P}_k$, representing the total accumulated hit count for the Gaussian across all views. α_k is the opacity, and $\gamma(\Sigma_k)$ is a volume-aware weighting factor based on Gaussian's spatial extent. The volume of the kth 3-D Gaussian is computed as $V(\mathbf{\Sigma}_k) = 4/3\pi abc$, where a, b, and c correspond to Gaussian's scaling factors along its three principal axes, obtained from the diagonal of its scale matrix $\boldsymbol{S}_k$.

While incorporating Gaussian volume helps suppress noisy or unstable primitives, directly using raw volume magnitudes may lead to over-pruning of small but important Gaussians that capture fine-scale geometry. To mitigate this issue, we adopt a normalized and bounded volume weighting strategy

$$\gamma(\mathbf{\Sigma}_k) = \left(\min\left(\frac{V(\mathbf{\Sigma}_k)}{V_{\max 90}}, 1\right)\right)^{\beta} \tag{8}$$

where $V_{\max 90}$ is the 90th percentile of all Gaussian volumes in the scene. This normalization bounds the weight within $(0, 1]$ and avoids overemphasizing large Gaussians, such as those associated with background regions. The exponent $\beta \in [0, 1]$ serves as a soft attenuation factor to control the sensitivity to volume differences.

Here, each component is designed with a specific physical intuition. The accumulated hit count $|\mathcal{P}_k|$ acts as a strong proxy for multiview geometric consistency. In satellite scenes with relatively constrained viewing angles, primitives that are consistently observed across multiple views are highly likely to correspond to stable 3-D structures, whereas artifacts like floaters are often view-dependent and thus have a lower hit count. The opacity α_k serves as a direct measure of the rendering contribution of the primitive. The volume utility function $\gamma(\mathbf{\Sigma}_k)$ is designed to explicitly model the contribution of a primitive's scale. A naive score that ignores the volume is inherently biased, as $|\mathcal{P}_k|$ is strongly correlated with scale.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

Our function addresses this by penalizing primitives that are too small while capping the influence of excessively large primitives.

Based on the significance scores GS_k, we rank all primitives in the descending order and retain only the top percent ξ for the subsequent stages. This primary pruning stage effectively removes the most obviously unstable or low-contribution Gaussians.

2) Opacity Entropy Regularization: While the coarse pruning described culls Gaussians with low overall contribution, it does not fully resolve geometric ambiguities that can arise during optimization. As astutely observed in recent work [56], the photometric loss alone is insufficient to tie primitives to a distinct surface. Semitransparent floater Gaussians can persist in empty space, as they can contribute to the final image color without being geometrically sound. This phenomenon is particularly detrimental for accurate surface reconstruction, introducing significant noise and artifacts into the final DSM.

Furthermore, this same opacity ambiguity can lead to local redundancy on the surface itself, where a single area is inefficiently modeled by multiple overlapping primitives instead of a single, decisive one. Both of these issues stem from the model's ability to exploit intermediate opacity values. To account for the predominantly opaque nature of target scenes in satellite photogrammetry (e.g., terrain and buildings) and encourage a more decisive and geometrically clean representation, we introduce a refinement step that leverages a binary entropy loss on the opacities

$$\mathcal{L}_{ent} = -\sum_{k} \left(\alpha_k \ln(\alpha_k) + (1-\alpha_k)\ln(1-\alpha_k)\right). \tag{9}$$

By minimizing this loss, we guide the model toward a state of minimal ambiguity. The entropy term actively penalizes semitransparent Gaussians, forcing them to decisively become either fully opaque as part of a coherent surface or fully transparent and thus geometrically irrelevant. This interaction provides an effective, continuous mechanism for model simplification. As the entropy loss reduces the opacity of unnecessary floater or redundant Gaussians, they are subsequently removed by the fixed opacity thresholding that runs throughout the training. The result is a highly compact and geometrically coherent set of primitives, which is essential for high-fidelity DSM extraction.

C. Uncertainty-Aware Learning

While our Geometry Refinement module effectively eliminates large-scale geometric artifacts, object surfaces in multitemporal satellite imagery can still exhibit photometric inconsistencies caused by illumination and shadow changes. As 3DGS attempts to learn a unified 3-D scene representation from images captured at different times and under varying lighting conditions, these discrepancies lead to ambiguities during the reconstruction process. For instance, if a single Gaussian kernel is forced to explain all illumination variations when rendering pixels from different time phases, it can distort the geometric structure, posing significant challenges for reconstruction. To address these fine-grained artifacts induced by temporal variations and further enhance the geometric accuracy, we introduce a UAL module.

Inspired by recent works that handle data inconsistencies in unconstrained photo collections [50], [57], [58], our method learns to identify and down-weight unreliable regions during optimization. The core idea behind our UAL module is to allow Gaussian kernels to accept certain rendering errors when encountering pixels from different time phases and lighting conditions during optimization. This effectively mitigates the impact of discrepancies arising from shadows and illumination changes. Specifically, we utilize a lightweight predictor to generate pixel-level confidence weights that modulate the main photometric loss 5. This predictor receives deep semantic features F extracted by a pretrained DINOv2 [59] encoder from the real image and predicts a per-pixel uncertainty value σ. It identifies unreliable regions largely affected by illumination or shadows. In such regions, simple color-based loss functions often fail; thus, the predictor effectively guides the optimization by focusing on more stable cues.

To theoretically justify the optimization of the predictor, we formulate the loss based on the modeling of heteroscedastic aleatoric uncertainty [60]. While original formulations applied this to photometric error [41], [60], we follow the recent WildGaussians approach [58] by adapting it to semantic features. The predictor's weights are optimized through a custom loss function $\mathcal{L}_{uncer}$

$$\mathcal{L}_{uncer} = \frac{\mathcal{L}_{dino}\left(\tilde{F}, F\right)}{2\sigma^2} + \lambda_{prior}\log\sigma + \left\| 1 - \frac{1}{2\sigma^2} \right\|_2^2 \tag{10}$$

where the first term is a data term that leverages a DINO-based feature similarity metric $\mathcal{L}_{dino}$ between rendered features $\tilde{F}$ and ground-truth features F

$$\mathcal{L}_{dino}\left(\tilde{F}, F\right) = \min\left(1, 2 - \frac{2\tilde{F}\cdot F}{\left\|\tilde{F}\right\|_2 \|F\|_2}\right). \tag{11}$$

This metric is robust to minor appearance changes but yields high values for significant feature discrepancies. Theoretically grounded in the modeling of heteroscedastic aleatoric uncertainty [60], the objective function in (10) effectively performs intelligent robust regression. Specifically, the first term uses the predicted uncertainty to adaptively weight the supervision signal, enabling the network to attenuate the impact of feature inconsistencies in unreliable regions. To counterbalance this and preclude the trivial solution where infinite uncertainty minimizes the global loss, the second term $\log\sigma$ acts as a regularization penalty. Consequently, the model is incentivized to predict high uncertainty only when the feature discrepancy is genuinely high. Furthermore, the third term functions as a soft prior constraint, biasing the raw confidence weights $1/2\sigma^2$ toward unity. This regularization is beneficial for stabilizing the optimization during the early training stages.

The predicted uncertainty σ is then used to generate a visibility map, which weights the confidence of each pixel

$$V_m = \min\left(\frac{1}{2\sigma^2}, 1\right). \tag{12}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

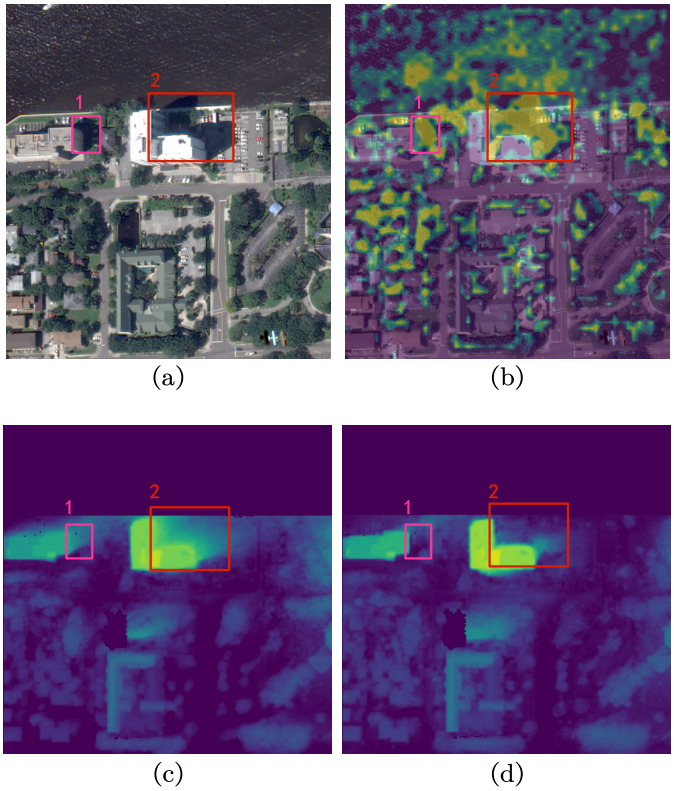


Fig. 2. Qualitative visualization of uncertainty on scene JAX_260. (a) Reference view. (b) Learned uncertainty mask ($1 - V_m$). (c) DSM generated by the baseline model. (d) DSM generated by the baseline with UAL (Base + UAL). In our method, V_m acts as the confidence weight derived from the predicted uncertainty σ. We visualize $1 - V_m$ instead of raw σ to clearly illustrate the effective weight reduction applied during optimization. Consequently, yellow regions indicate high uncertainty, effectively capturing photometric inconsistencies in shadows (boxes 1 and 2), vegetation, and highly reflective water surfaces. In the bottom row, we specifically highlight the shadow regions (boxes 1 and 2) to demonstrate how down-weighting these areas eliminates artifacts and restores clean surface geometry.

This visibility map is applied to both the rendered colors $\tilde{C}$ and ground-truth colors C via elementwise product $\odot$ to obtain their masked versions, $\tilde{C}_m = V_m \odot \tilde{C}$ and $C_m = V_m \odot C$.

The final reconstruction loss is then computed on these reliably reconstructed regions

$$\mathcal{L}_{render} = (1 - \lambda)\,\mathcal{L}_1\left(\tilde{C}_m, C_m\right) + \lambda \mathcal{L}_{\text{D-SSIM}}\left(\tilde{C}_m, C_m\right). \quad (13)$$

As shown in Fig. 2, the proposed UAL effectively mitigates the adverse impact of photometric inconsistencies caused by illumination changes, shadows, and reflective surfaces. By down-weighting these unreliable regions during optimization, the reconstruction process is guided to rely more on illumination-stable and geometrically consistent areas. As a result, the reconstructed DSM exhibits fewer artifacts and improved surface consistency, which is qualitatively demonstrated in the bottom row of Fig. 2. This improvement is further quantitatively validated by the mean absolute error (MAE) reduction reported in the experimental section. The complete training objective combines this weighted loss with previously defined regularization terms [(6) and (9)]

$$\mathcal{L}_{total} = \mathcal{L}_{render} + \lambda_{ent}\mathcal{L}_{ent} + \mathcal{L}_{reg}. \quad (14)$$

TABLE I
SPECIFICATIONS OF EXPERIMENTAL AOIS FROM DFC2019 AND IARPA2016 DATASETS

AOI Index	Train/Test	Alt. bounds [m]	Anchor (lat, lon)
JAX_004	9/2	[-24, 1]	(30.357, -81.706)
JAX_068	17/2	[-27, 30]	(30.348, -81.663)
JAX_214	21/3	[-29, 73]	(30.316, -81.663)
JAX_260	15/2	[-30, 13]	(30.311, -81.663)
JAX_264	21/3	[-29, 2]	(30.312, -81.653)
IARPA_001	25/4	[12, 59]	(-34.490, -58.584)
IARPA_002	21/3	[12, 80]	(-34.447, -58.575)
IARPA_003	21/2	[12, 80]	(-34.417, -58.575)

IV. EXPERIMENTS

In this section, we present the empirical validation of our method. We first detail the satellite imagery dataset and the quantitative metrics used for evaluation. Next, we provide the implementation specifics for our method and the SOTA approaches against which we compare. We then present a comprehensive analysis of the comparative results, and conclude with a series of ablation studies to validate the contribution of each key component in our method.

A. Datasets and Evaluation Metrics

We conduct our experiments on the datasets from the 2019 IEEE GRSS Data Fusion Contest (DFC2019) [21] and 2016 IARPA MVS 3-D Mapping Challenge (IARPA2016) [22]. Both datasets consist of multiview, multidate satellite imagery captured by the WorldView-3 satellite. The imagery features a spatial resolution of approximately 0.3 m. For evaluating the geometric accuracy of our 3-D reconstruction, we use the corresponding airborne LiDAR-derived DSMs as ground truth, which have a resolution of 0.5 m/pixel for DFC2019 and 0.3 m for IARPA2016. Building upon the experimental setup of the prior work [12], [18], we focus our evaluation on eight specific areas of interest (AOIs), including five scenes from DFC2019 Jacksonville collection and three from IARPA2016 dataset. These scenes were selected to provide a diverse range of complex terrains, including urban buildings, vegetation, and water bodies. Consistent with the benchmark specifications, the original high-resolution satellite images are cropped based on the geographical area of the corresponding DSM. This results in image crops of approximately 800×800 pixels for each view, covering a physical area of 256×256 m. Detailed specifications for each AOI, including the number of views for training and testing, altitude bounds, and geographic coordinates, are provided in Table I.

To quantitatively evaluate our method, we assess both the geometric accuracy of the reconstruction and the computational efficiency of the process. Following standard practice, our primary metric for geometric accuracy is the mean absolute error (MAE), which measures the average per-pixel L1 distance between the elevations in our generated DSM and the high-resolution, LiDAR-derived ground-truth DSM. The MAE is formulated as follows:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|h_i^{\text{pred}} - h_i^{\text{gt}}\right| \quad (15)$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE COMPARISON OF DSM GENERATION ACCURACY MAE. THE BEST RESULTS ARE IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method	JAX						IARPA				Time ↓
	004	068	214	260	264	Mean ↓	001	002	003	Mean ↓	
S2P [6]	2.155	1.701	2.816	2.314	1.273	2.052	3.365	4.649	2.983	3.666	**20min**
S-NeRF [14]	1.472	1.374	2.406	2.299	1.754	1.852	3.649	4.907	3.693	4.083	20h
Sat-NeRF [12]	**1.288**	1.249	2.009	1.864	1.694	1.621	1.650	2.060	2.934	2.215	20h
SAT-NGP [45]	1.478	1.393	1.978	1.860	3.644	2.071	–	–	–	–	**20min**
EO-NeRF [15]	1.406	1.590	2.580	1.948	1.096	1.724	1.624	**1.942**	2.109	1.892	6h
Sat-DN [47]	1.736	**1.063**	1.931	**1.585**	1.396	1.542	–	–	–	–	3.5h
EOGS [18]	2.083	1.986	2.854	3.523	1.200	2.329	1.588	2.005	**2.048**	1.880	55min
Ours	1.982	1.173	**1.821**	1.758	**0.820**	**1.511**	**1.546**	1.968	2.053	**1.856**	30min

TABLE III
QUANTITATIVE COMPARISON RESULTS OF ABLATION EXPERIMENTS ON JAX. THE BEST RESULTS ARE IN BOLD

	004	068	214	260	264
Base	2.083	1.986	2.854	3.523	1.200
Base + GR	2.079	1.288	1.942	2.179	0.926
Base + UAL	2.006	1.647	2.368	2.336	0.851
Base + GR + UAL	**1.982**	**1.172**	**1.821**	**1.758**	**0.820**

where N is the total number of pixels in the evaluated area, h_i^{pred} is the predicted elevation at pixel i, and h_i^{gt} is the corresponding ground-truth elevation.

In addition to accuracy, we consider computational efficiency to be a critical factor for practical applications, especially given the growing scale and volume of satellite image archives. Therefore, we also report the average training time required for our method to converge for each scene.

B. Comparison Methods

1) S2P: As a stereovision baseline, we compare against S2P [6], a well-established satellite stereo pipeline for 3-D reconstruction that won the 2016 IARPA MVS 3-D Mapping Challenge. This method uses the MGM method to process ten manually selected stereo image pairs. The individual point clouds generated from each pair are merged to form a complete scene representation, which is then processed to generate the final DSM. Any remaining holes are filled using the inverse distance weighting.

2) S-NeRF and Sat-NeRF: S-NeRF [14] is the first NeRF variant for satellite imagery that models direct and indirect illumination induced by the sun and sky. Sat-NeRF [12] is a subsequent development that extends S-NeRF by explicitly modeling shadows and incorporating a mechanism to handle transient objects.

3) Sat-NGP: To evaluate performance against modern accelerated NeRF architectures, we also compare our method with SAT-NGP [45]. This work drastically reduces training time by using a multiresolution hash encoding for scene representation.

4) EO-NeRF: EO-NeRF [15] enforces geometrically consistent shadow rendering by casting solar rays from the surface, ensuring that shadows strictly align with the geometry. In addition, it also integrates internal bundle adjustment to refine camera parameters during training.

5) Sat-DN: Sat-DN [47] is a recently proposed implicit surface reconstruction framework leveraging multiresolution hash grids. To address weak textures and illumination inconsistencies, it incorporates explicit geometric guidance derived from monocular depth priors and surface normal consistency constraints, along with a progressive training strategy.

6) EOGS: We include EOGS [18] as a key baseline, as it is the first method to specifically adapt 3DGS for digital elevation modeling. Its main contributions include using an affine camera approximation for satellite sensors and introducing specific regularization terms to improve the geometric quality. The method is notable for achieving accuracy comparable with NeRF-based approaches with significantly reduced training times.

C. Implementation Details

Our method is implemented on the basis of the EOGS [18] code base, and we adopt its initialization strategy, where the scene is populated with Gaussian primitives at a uniform spatial density of 0.13 Gaussians/m^3, with each primitive initialized to a white color and a minimal opacity of 0.01. Our method is optimized over 10 000 training iterations. For our proposed method, we configure the parameters as follows. The significance-guided pruning is performed once at the 5000 iteration, where we retain the top 60% of the most significant Gaussians. The volume-aware weighting factor in the significance score is controlled by setting β to 0.5. Immediately following this pruning step, we introduce the opacity entropy regularization for the remainder of the training, with its weight λ_{ent} linearly increased from 10^{-5} to 10^{-4} over the next 2000 iterations. Our uncertainty modeling relies on features from the DINOv2-S. The lightweight uncertainty predictor is trained from the start of the optimization process using the Adam optimizer [61] with a learning rate of 0.001. The prior regularization term to prevent σ from being infinitely large is weighted by $\lambda_{prior} = 0.5$. To ensure training stability, the application of the uncertainty mask is delayed and gradually introduced. The masking of the main photometric loss starts at iteration 1000, with its weight being linearly ramped up

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

to its final value over a 500-iteration warm-up period. Unless otherwise specified, all other hyperparameters and the Adam optimization settings for the Gaussian attributes follow the original 3DGS implementation.

For baseline comparisons, we established a rigorous evaluation pipeline. For S-NeRF and Sat-NeRF, the authors provided pretrained models for four of our five test scenes. We utilized these models and verified that our evaluation process successfully reproduced the exact performance metrics reported in their original publications [12]. This successful verification provides high confidence in our experimental setup. For the fifth scene, where pretrained models were unavailable, and for all other comparative methods, we utilized their public codebases to train and test the models under unified conditions. Results for SAT-NGP and Sat-DN on the IARPA2016 dataset are denoted by "–," as their implementations could not be successfully adapted to converge on this dataset under the unified experimental settings. For the DFC2019 dataset, while the original EOGS paper demonstrated strong performance on custom-processed, high-dynamic-range pansharpened data, our evaluation is conducted on the more standard 8-bit RGB images. This approach ensures a fair and direct comparison of all benchmarked methods under consistent conditions. Therefore, the performance metrics reported for EOGS are derived from our own implementation, trained for 10 000 iterations under the same conditions as our proposed method. For the IARPA2016 dataset, we follow the original EOGS setting using raw pansharpened inputs. During the evaluation phase, semantic masking is applied, where water masks are available, and elevation outliers are clamped to within a 10-m margin of the ground truth to ensure the robust metric calculation.

All our experiments were conducted on Ubuntu 22.04 system with a hardware environment consisting of an AMD Ryzen 7 5700X CPU, 64-GB RAM, and a single NVIDIA RTX 4090 GPU with 24-GB VRAM.

D. Performance Comparison and Analysis

The performance of our proposed method, GU-GS, was comprehensively evaluated and compared with SOTA methods on both the DFC2019 and IARPA2016 datasets. We present both qualitative and quantitative results, which consistently demonstrate the superiority of our approach. Qualitatively, as shown in the visual comparisons in Figs. 3 and 4, our method produces DSMs with significantly higher fidelity, sharper geometric details, and fewer artifacts than competing methods. These visual improvements are strongly supported by the quantitative metrics in Table II, where GU-GS achieves the best overall geometric accuracy among all methods. A detailed analysis reveals the distinct limitations of the prior work and underscores the advantages of our approach.

We begin our analysis with the traditional stereovision pipeline, S2P. While this method is capable of producing complete surfaces without holes, a detailed inspection across all scenes reveals that it consistently suffers from significant oversmoothing, which leads to a critical loss of geometric accuracy and fine details. This limitation is most pronounced in scenes with dense small structures. In JAX_004, distinct small-scale buildings present in the LiDAR ground truth are merged into a single, indiscernible mass. Similarly, in IARPA_001, the method fails to resolve the complex internal courtyard structures, fusing separate building components into a solid block. For structures with sharp geometric features, such as the "X"-shaped building in JAX_068 and the "L"-shaped building in JAX_260, S2P fails to preserve clear structural boundaries, resulting in a distorted, blurred appearance. This issue is further exemplified in IARPA_002, where the corners of the cross-shaped buildings are overly smoothed into rounded shapes. Furthermore, the method struggles to reconstruct planar surfaces, producing geometric noise on rooftops in scenes like JAX_214. In IARPA_003, this manifests as distorted edges and irregular undulations on the large flat roof. These characteristic failures stem from the inherent limitations of local patch-based matching in texture-poor regions and the heavy smoothing regularization required by traditional stereo pipelines to produce dense outputs. Consequently, while S2P serves as an important baseline, its performance underscores the necessity of modern learning-based approaches for high-fidelity reconstruction.

Moving to the learning-based approaches, we first analyze the results from the NeRF-based and implicit methods, including early variants (S-NeRF and Sat-NeRF) and recent accelerated or guided approaches (SAT-NGP, EO-NeRF, and Sat-DN). Our analysis reveals that while recent advancements improve performance, these methods are highly scene-dependent and often suffer from a shared, systemic drawback: a susceptibility to geometric noise and artifacts stemming from their implicit volumetric representation. The severity of these issues varies with scene characteristics. For instance, in JAX_068, which features strong textures and significant parallax, all NeRF-based methods produce a reasonably coherent macrostructure, though they still exhibit surface irregularities and a loss of fine detail upon close inspection. However, their performance degrades significantly in less constrained environments. In scenes like JAX_004 and JAX_260, which contain areas of sparse features, these methods produce pervasive salt-and-pepper noise that obscures the underlying geometry. In JAX_260, the lack of strong texture near the building appears to cause the volumetric density to diffuse at object boundaries, resulting in the indistinct and blurry outlines observed in the Sat-NeRF and EO-NeRF results. In JAX_214, while the macro structure of buildings is recognizable, the rooftops are not reconstructed as clean and consistent surfaces. Instead, they are mottled with erroneous geometric fluctuations, creating bumpy and unrealistic textures. A similar failure mode is evident in IARPA_003, where the large industrial rooftops exhibit significant distortions rather than a consistent geometric plane. The most significant failure mode is observed over the non-Lambertian water surface in JAX_264. The failure of SAT-NGP is particularly catastrophic on the JAX_264 water surface, resulting in chaotic and entirely erroneous geometry. This visual failure corresponds to an extremely high MAE of 3.644 m. Crucially, even Sat-DN, which utilizes depth and normal supervision, struggles to reconstruct an accurate water surface. This is likely due to their hash-grid representation struggling to converge in the absence of a consistent photometric signal from the textureless

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

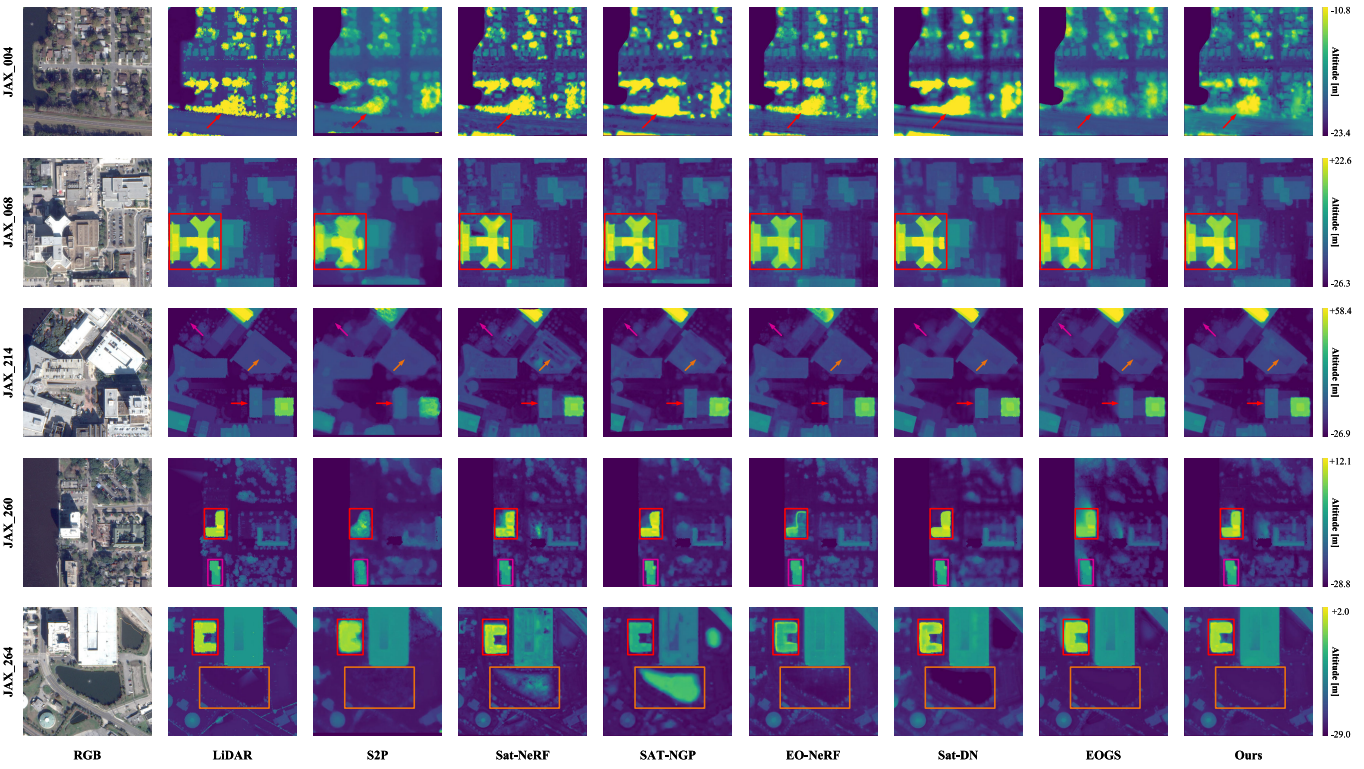


Fig. 3. Qualitative results of DSMs generated from JAX. Each row displays the results for a single scene from all competing methods, while each column tracks the performance of one method across the different test scenes. Areas with significant differences are highlighted with bounding boxes, while arrows are used to indicate specific discrepancies in irregular regions unsuitable for boxes. An altitude legend is provided on the right for reference. For visual context, the leftmost column provides a corresponding RGB image of each scene from a near-nadir perspective.

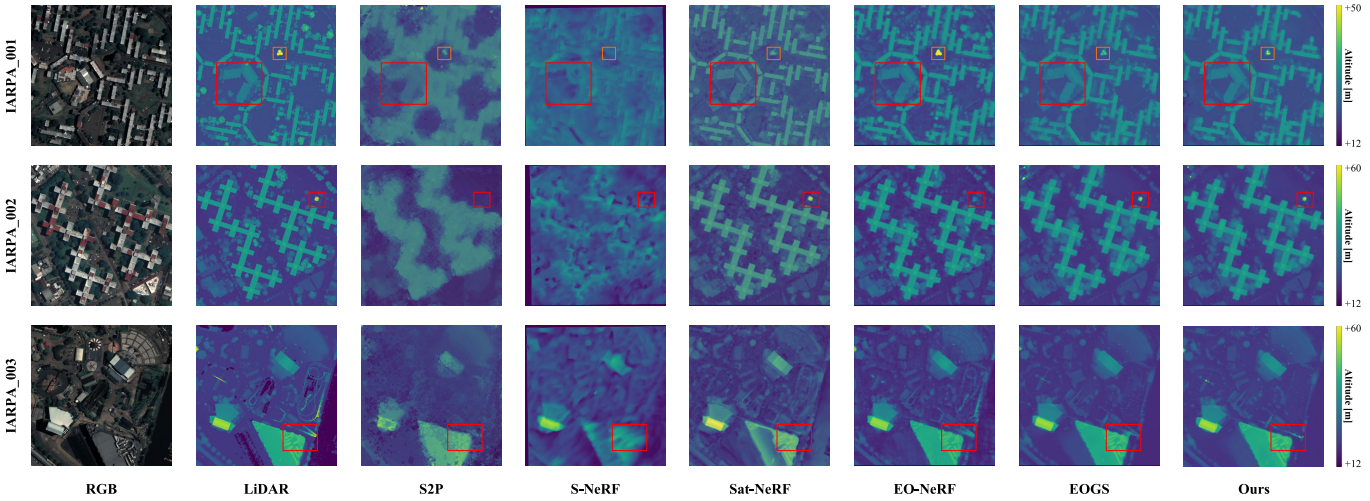


Fig. 4. Qualitative results of DSMs generated from IARPA. Each row displays the results for a single scene from all competing methods, while each column tracks the performance of one method across the different test scenes. Areas with significant differences are highlighted with bounding boxes. An altitude legend is provided on the right for reference. For visual context, the leftmost column provides a corresponding RGB image of each scene from a near-nadir perspective.

and reflective water surface, highlighting the vulnerability of such accelerated methods to challenging surface types. These findings indicate that while NeRF-based methods can succeed under ideal conditions, as observed in IARPA_001 and IARPA_002, their reliance on photometric consistency from a volumetric density field makes them fundamentally unreliable for producing clean, accurate DSMs across diverse and challenging satellite scenes. Furthermore, these methods generally suffer from low efficiency.

As the key 3DGS-based baseline for satellite photogrammetry, EOGS represents a significant leap in efficiency over NeRF-based methods. However, its optimization process, lacking targeted geometry refinement, still struggles to produce clean surfaces from the challenging multitemporal satellite

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

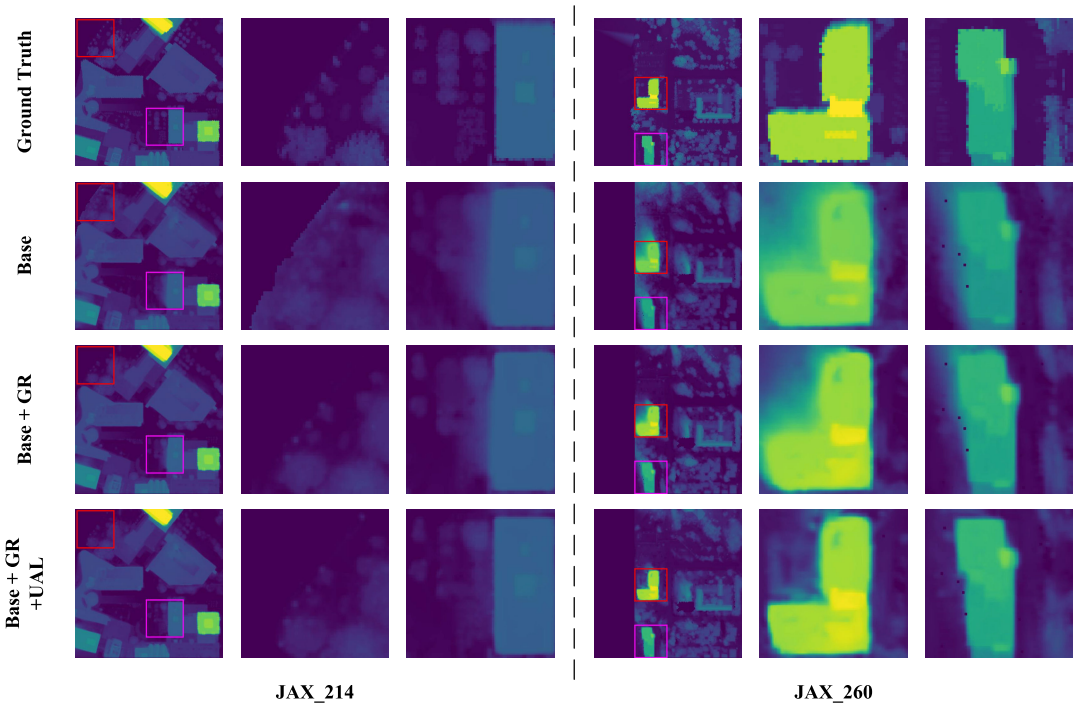


Fig. 5. Qualitative results of our ablation study on two representative scenes, JAX_214 and JAX_260. Each row visualizes the progressive improvement from our baseline (Base) to our full model (Base + GR + UAL). Within each result block, (left) full DSM, followed by two zoomed-in regions of interest. These two scenes were selected to demonstrate the complementary roles of our proposed components in addressing different types of geometric challenges. In JAX_214, the GR module (+GR) first cleans major artifacts from the complex building surfaces, after which the UAL module (+UAL) removes residual surface noise. Similarly, in JAX_260, our method first eliminates significant floating artifacts from the scene and then reduces fine-grained noise in the background.

data. The EOGS results are consistently degraded by the geometric noise stemming from opacity ambiguity, where numerous semitransparent primitives persist in empty space. This issue manifests visually as blurry boundaries and a hazy layer of floating artifacts around artificial structures. As seen in JAX_068, JAX_260, and JAX_264, the edges of buildings are poorly defined. This systemic limitation is also evident in IARPA_003, where the hazy artifacts cause building edges to appear fused with adjacent structures, obscuring the clear separation present in the ground truth. This is particularly severe in JAX_260, where the DSM is corrupted by pervasive noise, leading to a very high MAE of 3.523 m. Furthermore, EOGS struggles to model complex natural features. In JAX_214, the method produces erroneous ground elevation estimates near the water-land boundary in the top-left region. This is likely caused by the model's inability to accurately represent the complex geometry of the adjacent vegetated areas, leading to artifacts that spill over and corrupt the ground plane. Furthermore, in IARPA_001 and IARPA_002, the method fails to capture fine-scale geometry; the distinct small building highlighted in the box suffers from height underestimation, appearing faint and flattened compared with the LiDAR ground truth. These limitations highlight that a direct application of 3DGS to this task is insufficient for generating high-fidelity DSMs.

Our GU-GS method addresses the aforementioned deficiencies, demonstrating a robust and efficient path to high-fidelity reconstruction. The visual results in Figs. 3 and 4, supported by the quantitative metrics in Table II, show a consistent and significant improvement over other methods. Our method achieves the best overall mean MAE of 1.511 m on the JAX dataset and 1.856 m on the IARPA dataset, outperforming competitors in terms of average accuracy. The most significant improvement is observed in the challenging JAX_264 scene, which contains a large water body. Here, GU-GS achieves an exceptionally low MAE of 0.820 m, representing a 25.2% error reduction compared to the next-best method EO-NeRF. Crucially, this top-tier accuracy is not achieved at the expense of efficiency. As shown in Table II, GU-GS requires only 30 min of training per scene. This represents not only a dramatic ~40× speedup compared with the ~20 h required by leading NeRF-based methods but is also achieved in nearly half the time of the direct 3DGS baseline, all while delivering substantially higher accuracy, making our method a more practical and scalable solution for large-scale applications. While, like other methods, challenges remain in perfectly reconstructing the extremely high-frequency details of JAX_004 or small isolated structures in IARPA_002, our method still demonstrates a clear advantage over the EOGS baseline by effectively suppressing the most severe noise artifacts. In summary, GU-GS demonstrates an excellent balance of accuracy and efficiency, achieving highly competitive performance for this task.

E. Ablation Studies

To rigorously validate the contribution of each key component in our GU-GS method, we conduct a series of ablation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

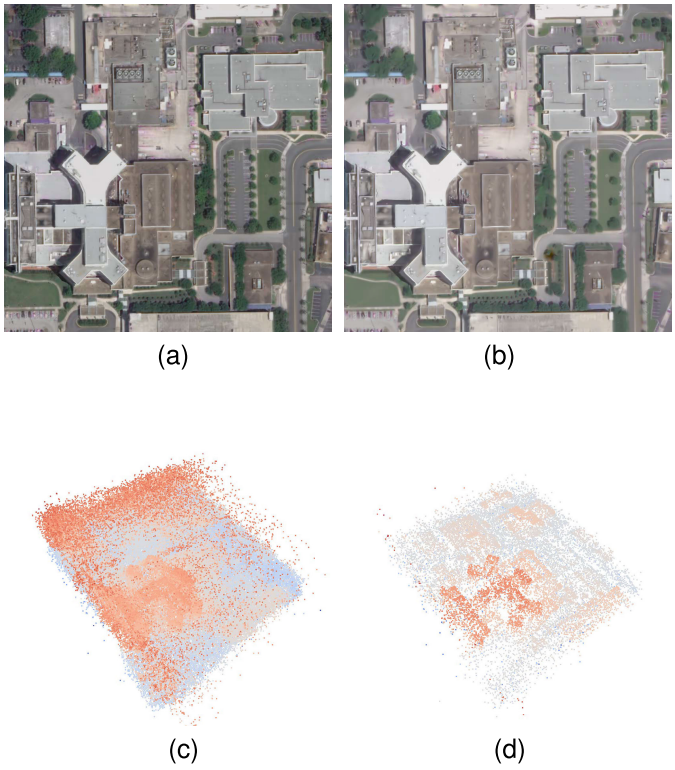


Fig. 6. Qualitative analysis of our GR module on scene JAX_068. (Top row) Final rendered images from (a) the base model and (b) our refined model. (Bottom row) Visualizes the corresponding Gaussian primitives (c) before and (d) after GR. Despite a significant reduction in primitives, the render quality is visually indistinguishable.

TABLE IV
NUMBER OF GAUSSIAN PRIMITIVES ON JAX

	004	068	214	260	264
Base	131K	153K	262K	122K	68K
Base + GR	7K	19K	24K	11K	8K
Base + GR + UAL	6K	16K	20K	8K	7K

studies on all five JAX scenes. We begin with a baseline model (Base) and progressively integrate our geometry refinement (GR) module and the UAL module. Fig. 5 provides a qualitative overview of these progressive improvements on two representative scenes, JAX_214 and JAX_260. The quantitative results are summarized in Table III, with detailed qualitative visualizations provided in Figs. 6–8.

1) Geometry Refinement: The base model, relying on the standard EOGS framework, often produces geometrically noisy results, as it lacks a mechanism to explicitly remove redundant or unstable primitives. This is evident in the Base row of Fig. 5, where both JAX_214 and JAX_260 scenes exhibit significant floating artifacts and blurred building edges. Our first contribution, the geometry refinement module, is designed to resolve these large-scale geometric errors.

Table IV first demonstrates the module's powerful compression capability, reducing the number of Gaussian primitives by $\sim 10\times$ on average. Fig. 6 provides a direct visualization of this process for the same scene. The bottom row of Fig. 6(c) and (d) clearly illustrates the removal of a massive

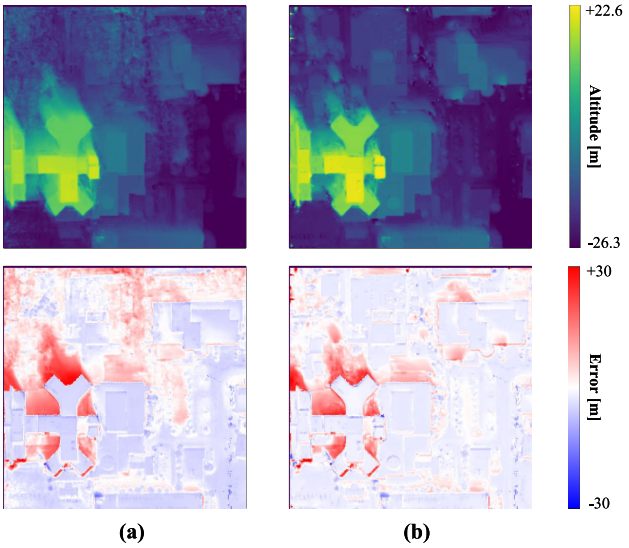


Fig. 7. Qualitative analysis of the GR module under a five-view training setting. (Top row) Compares the generated DSMs from (a) baseline model and (b) our model with GR module. (Bottom row) Visualizes their respective error maps against the ground truth. Our method significantly suppresses the floating artifacts and refines the geometric details. Both DSMs and error maps are rendered with shared colorbars for fair comparison.

number of floating noise primitives, while the top row of Fig. 6(c) and (d) confirms that this significant reduction in complexity has no discernible negative impact on the final novel view synthesis quality, validating the efficiency and quality-preserving nature of our pruning strategy. This targeted refinement of the underlying geometry translates directly into a dramatic improvement in DSM accuracy. As shown in Table III, adding the refinement module (Base + GR) consistently and significantly lowers the MAE across all scenes, with the error for JAX_068 improving by over 35% (from 1.986 to 1.288 m). The qualitative results in Fig. 6 confirm that this improvement is due to the successful elimination of floaters and the refinement of geometric boundaries.

To further investigate why our geometry refinement module is so effective, we conduct a challenging few-shot experiment using only the first 5 images from the JAX_068 sequence. Fig. 7 provides a qualitative comparison. The top row shows that the baseline model, as shown in Fig. 7(a), produces a blurry DSM corrupted by significant noise, resulting in an MAE of 4.33 m. This is further confirmed by the large, pervasive positive errors, displayed as red regions, in its corresponding error map (bottom row). These pervasive positive errors and blurry DSMs are a direct consequence of the limitations of standard 3DGS in poorly constrained satellite imaging. Under severe multiview shortages, the optimization generates numerous unstable or floating Gaussians, or many semitransparent ones at true surfaces. As these Gaussians are often closer to the camera, they prematurely accumulate opacity weights $T_k\alpha_k$ along the viewing ray, erroneously lifting the DSM height and blurring surface boundaries. In stark contrast, our method with the GR module, as shown in Fig. 7(b), yields a much sharper and cleaner DSM, significantly reducing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

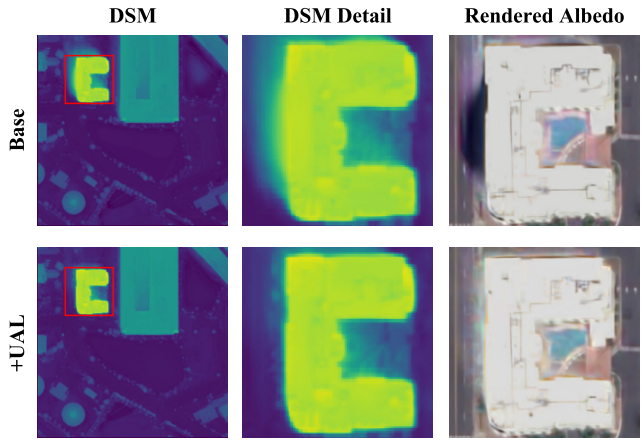


Fig. 8. Qualitative analysis of our UAL module on scene JAX_264. This figure compares the results from (top row) the baseline model against (bottom row) the baseline augmented with our UAL module to isolate its contribution. The module's effectiveness is twofold. First, it enhances geometric accuracy, as seen in the DSM detail column, where the building edges in the bottom row are significantly sharper than in the top row. Second, it corrects major rendering artifacts. In the rendered albedo column, the large, dark artifact present on the building facade in the baseline result is successfully removed after applying the uncertainty module.

the MAE to 2.97 m. The corresponding error map shows a dramatic reduction in large-scale positive errors, demonstrating the module's remarkable robustness. This is because the accumulated hit count $|\mathcal{P}_k|$ in our significance score acts as a powerful regularizer for multiview consistency. It effectively prunes primitives that are not consistently observed across the limited views, preserving only geometrically stable structures. Concurrently, opacity entropy regularization drives remaining Gaussians toward binary opacity states, effectively diminishing the weight of ambiguous Gaussians. This experiment highlights that our pruning strategy is a principled approach to enforcing geometric consistency, especially in poorly constrained scenarios.

2) Uncertainty-Aware Learning: While the geometry refinement module addresses large scale, internal model redundancy, the UAL module is designed to tackle the more subtle problem of external data corruption from view inconsistencies in the multitemporal imagery. This component performs a fine-grained refinement of both the geometry and the rendered appearance. The effectiveness of this module is demonstrated quantitatively in Table III. The addition of the uncertainty module (Base + GR + UAL) yields a consistent reduction in MAE across all scenes, enabling our full model to achieve its SOTA performance.

Fig. 8 provides a detailed qualitative analysis of the module's impact on scene JAX_264. Its contribution is twofold. First, it enhances the final DSM accuracy. As seen in the "DSM Detail" column, the building edges in the result from our full model (bottom row) are visibly sharper and more clearly defined than in the baseline (top row). Second, and more significantly, the module corrects major rendering artifacts caused by data inconsistencies. This is best observed in the "rendered albedo" column. The rendered albedo should represent the intrinsic surface color, independent of lighting phenomena such as cast shadows. However, the baseline result (top row) exhibits a large, dark artifact on the building facade, which erroneously conflates lighting effects with surface color. Our method (bottom row) successfully removes this artifact, producing a clean and coherent albedo that correctly represents the underlying material. This demonstrates the module's crucial role in regulating the training process by identifying and down-weighing unreliable pixel information, leading to improvements in both geometric fidelity and visual quality.

V. Conclusion

In this article, we introduced GU-GS, a novel method for refining 3DGS to achieve accurate and efficient DSM generation from multitemporal satellite imagery. Our work addresses the key limitations of applying standard 3DGS to this task, namely the prevalence of geometric artifacts and the sensitivity to photometric inconsistencies and shadow changes. We proposed a geometry refinement module to create a compact and coherent 3-D representation, effectively reducing redundant Gaussians and ambiguous surfaces. Furthermore, we introduced a UAL module to mitigate the impact of multiview inconsistent data, thereby enhancing the stability and accuracy of the optimization.

Experiments on the DFC2019 and IARPA2016 datasets validated the effectiveness of our approach. GU-GS establishes a new SOTA in geometric accuracy, outperforming traditional and recent learning-based methods, while being significantly more efficient than NeRF-based alternatives. Despite the strong performance, challenges remain in perfectly reconstructing extremely high-frequency details in scenes dominated by complex vegetation. Future work could explore the integration of structural priors to better handle such scenarios. Nevertheless, GU-GS provides an accurate and practical solution for satellite-based 3-D reconstruction.

Acknowledgment

The authors sincerely thank the contributors of the DFC2019 and IARPA2016 datasets for their support in this research, and Shuyu Hu, Tong Jin, Junze Shi, and Yang Lu for their discussion and support.

References

[1] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogramm. Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.

[2] A. Wallner et al., "A remote sensing-guided forest inventory concept using multispectral 3D and height information from ZiYuan-3 satellite data," *Forestry: Int. J. Forest Res.*, vol. 95, no. 3, pp. 1–16, May 2022.

[3] J.-Y. Rau, J.-P. Jhan, and Y.-C. Hsu, "Analysis of oblique aerial images for land cover and point cloud classification in an urban environment," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1304–1319, Mar. 2015.

[4] B. Neupane, T. Horanont, and J. Aryal, "Deep learning-based semantic segmentation of urban features in satellite images: A review and meta-analysis," *Remote Sens.*, vol. 13, no. 4, p. 808, Feb. 2021.

[5] L. Zhao, H. Wang, Y. Zhu, and M. Song, "A review of 3D reconstruction from high-resolution urban satellite images," *Int. J. Remote Sens.*, vol. 44, no. 2, pp. 713–748, Jan. 2023.

[6] C. de Franchis, E. Meinhardt-Llopis, J. Michel, J.-M. Morel, and G. Facciolo, "An automatic and modular stereo pipeline for pushbroom images," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 49–56, Aug. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

[7] J. Michel et al., "A new satellite imagery stereo pipeline designed for scalability, robustness and performance," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 171–178, Aug. 2020.

[8] H. Hirschmuller, "Stereo processing by semiglobal matching and mutual information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 30, no. 2, pp. 328–341, Feb. 2008.

[9] R. A. Beyer, O. Alexandrov, and S. McMichael, "The Ames stereo pipeline: NASA's open source software for deriving and processing terrain data," *Earth Space Sci.*, vol. 5, no. 9, pp. 537–548, Sep. 2018.

[10] R. Marí, T. Ehret, and G. Facciolo, "Disparity estimation networks for aerial and high-resolution satellite images: A review," *Image Process. Line*, vol. 12, pp. 501–526, Nov. 2022.

[11] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, no. 1, pp. 99–106, Dec. 2021.

[12] R. Marí, G. Facciolo, and T. Ehret, "Sat-NeRF: Learning multi-view satellite photogrammetry with transient objects and shadow modeling using RPC cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1310–1320.

[13] J. T. Barron, B. Mildenhall, M. Tancik, P. Hedman, R. Martin-Brualla, and P. P. Srinivasan, "Mip-NeRF: A multiscale representation for anti-aliasing neural radiance fields," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5835–5844.

[14] D. Derksen and D. Izzo, "Shadow neural radiance fields for multi-view satellite photogrammetry," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1152–1161.

[15] R. Marí, G. Facciolo, and T. Ehret, "Multi-date Earth observation NeRF: The detail is in the shadows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2035–2045.

[16] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," *ACM Trans. Graph.*, vol. 42, no. 4, pp. 1–14, Jul. 2023.

[17] Y. Liu, C. Luo, L. Fan, N. Wang, J. Peng, and Z. Zhang, "CityGaussian: Real-time high-quality large-scale scene rendering with Gaussians," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2025, pp. 265–282.

[18] L. S. Aira, G. Facciolo, and T. Ehret, "Gaussian splatting for efficient satellite image photogrammetry," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 5959–5969.

[19] J. Qian et al., "C3DGS: Compressing 3D Gaussian model for surface reconstruction of large-scale scenes based on multiview UAV images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4396–4409, 2025.

[20] M. Bosch, K. Foster, G. Christie, S. Wang, G. D. Hager, and M. Brown, "Semantic stereo for incidental satellite images," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1524–1532.

[21] B. Le Saux, N. Yokoya, R. Hansch, M. Brown, and G. Hager, "2019 data fusion contest [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 103–105, Mar. 2019.

[22] M. Bosch, Z. Kurtz, S. Hagstrom, and M. Brown, "A multiple view stereo benchmark for satellite imagery," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, Oct. 2016, pp. 1–9.

[23] G. Facciolo, C. D. Franchis, and E. Meinhardt, "MGM: A significantly more global matching for stereovision," in *Proc. Brit. Mach. Vis. Conf.*, Swansea, U.K., B. Press, Ed., Durham, U.K.: BMVA Press, 2015, pp. 1–13.

[24] L. Lastilla, R. Ravanelli, F. Fratarcangeli, M. Di Rita, A. Nascetti, and M. Crespi, "FOSS4G DATE for DSM generation: Sensitivity analysis of the semi-global block matching parameters," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 67–72, Jun. 2019.

[25] M. Rothermel, K. Wenzel, D. Fritsch, and N. Haala, "SURE: Photogrammetric surface reconstruction from imagery," Univ. Stuttgart, Stuttgart, Germany, Tech. Rep., 2013.

[26] J. Liu and S. Ji, "A novel recurrent encoder–decoder structure for large-scale multi-view stereo reconstruction from an open aerial dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 6049–6058.

[27] K. Bittner, L. Liebel, M. Körner, and P. Reinartz, "Long-short skip connections in deep neural networks for DSM refinement," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 383–390, Aug. 2020.

[28] F. Zhang, V. Prisacariu, R. Yang, and P. H. S. Torr, "GA-Net: Guided aggregation net for end-to-end stereo matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 185–194.

[29] A. Gómez, G. Randall, G. Facciolo, and R. G. von Gioi, "An experimental comparison of multi-view stereo approaches on satellite images," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 707–716.

[30] S. He, R. Zhou, S. Li, S. Jiang, and W. Jiang, "Disparity estimation of high-resolution remote sensing images with dual-scale matching network," *Remote Sens.*, vol. 13, no. 24, p. 5050, Dec. 2021.

[31] F. Zhang, X. Qi, R. Yang, V. Prisacariu, B. Wah, and P. Torr, "Domain-invariant stereo matching networks," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Berlin, Germany: Springer-Verlag, 2020, pp. 420–439.

[32] S. He, S. Li, S. Jiang, and W. Jiang, "HMSM-Net: Hierarchical multi-scale matching network for disparity estimation of high-resolution satellite stereo images," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 314–330, Jun. 2022.

[33] Z. Rao, M. He, Z. Zhu, Y. Dai, and R. He, "Bidirectional guided attention network for 3-D semantic detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6138–6153, Jul. 2021.

[34] J. L. Schönberger, E. Zheng, J.-M. Frahm, and M. Pollefeys, "Pixelwise view selection for unstructured multi-view stereo," in *Computer Vision–(ECCV)*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds., Cham, Switzerland: Springer, 2016, pp. 501–518.

[35] C. Stucker and K. Schindler, "ResDepth: A deep residual prior for 3D reconstruction from high-resolution satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 560–580, Jan. 2022.

[36] Q. Xu, W. Kong, W. Tao, and M. Pollefeys, "Multi-scale geometric consistency guided and planar prior assisted multi-view stereo," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4945–4963, Apr. 2023.

[37] K. Zhang, N. Snavely, and J. Sun, "Leveraging vision reconstruction pipelines for satellite imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 2139–2148.

[38] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4104–4113.

[39] J. Gao, J. Liu, and S. Ji, "A general deep learning based framework for 3D reconstruction from multi-view stereo satellite images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 446–461, Jan. 2023.

[40] K. Gao, Y. Gao, H. He, D. Lu, L. Xu, and J. Li, "NeRF: Neural radiance field in 3D vision: A comprehensive review (updated post-Gaussian splatting)," 2022, *arXiv:2210.00379*.

[41] R. Martin-Brualla, N. Radwan, M. S. M. Sajjadi, J. T. Barron, A. Dosovitskiy, and D. Duckworth, "NeRF in the wild: Neural radiance fields for unconstrained photo collections," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7206–7215.

[42] R. Cipolla, Y. Gal, and A. Kendall, "Multi-task learning using uncertainty to weigh losses for scene geometry and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7482–7491.

[43] S. Xie, L. Zhang, G. Jeon, and X. Yang, "Remote sensing neural radiance fields for multi-view satellite photogrammetry," *Remote Sens.*, vol. 15, no. 15, p. 3808, Jul. 2023.

[44] Y. Qu and F. Deng, "Sat-mesh: Learning neural implicit surfaces for multi-view satellite reconstruction," *Remote Sens.*, vol. 15, no. 17, p. 4297, Aug. 2023.

[45] C. Billouard, D. Derksen, E. Sarrazin, and B. Vallet, "SAT-NGP: Unleashing neural graphics primitives for fast relightable transient-free 3D reconstruction from satellite imagery," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 8749–8753.

[46] C. Zhang, Y. Yan, C. Zhao, N. Su, and W. Zhou, "FVMD-ISRe: 3-D reconstruction from few-view multidate satellite images based on the implicit surface representation of neural radiance fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624814.

[47] T. Liu, S. Zhao, W. Jiang, and B. Guo, "Sat-DN: Implicit surface reconstruction from multiview satellite images with depth and normal supervision," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 25374–25390, 2025.

[48] T. Wu et al., "Recent advances in 3D Gaussian splatting," *Comput. Vis. Media*, vol. 10, no. 4, pp. 613–642, Aug. 2024.

[49] G. Chen and W. Wang, "A survey on 3D Gaussian splatting," 2024, *arXiv:2401.03890*.

[50] N. Bai, A. Yang, H. Chen, and C. Du, "SatGS: Remote sensing novel view synthesis using multi- temporal satellite images with appearance-adaptive 3DGS," *Remote Sens.*, vol. 17, no. 9, p. 1609, May 2025.

[51] X. Huang et al., "SkySplat: Generalizable 3D Gaussian splatting from multi-temporal sparse satellite images," 2025, *arXiv:2508.09479*.

[52] P. Papantonakis, G. Kopanas, B. Kerbl, A. Lanvin, and G. Drettakis, "Reducing the memory footprint of 3D Gaussian splatting," *Proc. ACM Comput. Graph. Interact. Techn.*, vol. 7, no. 1, pp. 1–17, May 2024.

[53] L. Fan, Y. Yang, M. Li, H. Li, and Z. Zhang, "Trim 3D Gaussian splatting for accurate geometry representation," 2024, *arXiv:2406.07499*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.

[54] Z. Fan, K. Wang, K. Wen, Z. Zhu, D. Xu, and Z. Wang, "LightGaussian: Unbounded 3D Gaussian compression with 15x reduction and 200+ FPS," 2023, *arXiv:2311.17245*.
[55] R. Yang et al., "Spectrally pruned Gaussian fields with neural compensation," 2024, *arXiv:2405.00676*.
[56] B. Xu, J. Hu, J. Li, and Y. He, "GSurf: 3D reconstruction via signed distance fields with direct Gaussian supervision," 2024, *arXiv:2411.15723*.
[57] W. Ren, Z. Zhu, B. Sun, J. Chen, M. Pollefeys, and S. Peng, "NeRF on-the-go: Exploiting uncertainty for distractor-free NeRFs in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 8931–8940.
[58] J. Kulhanek, S. Peng, Z. Kukelova, M. Pollefeys, and T. Sattler, "WildGaussians: 3D Gaussian splatting in the wild," in *Proc. 38th Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates, 2025.
[59] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, vol. 2024, pp. 1–32, Jan. 2024.
[60] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.* Red Hook, NY, USA: Curran Associates, 2017, pp. 5580–5590.
[61] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Y. Bengio and Y. LeCun, Eds., San Diego, CA, USA, 2015, pp. 1–15.



Bin Ding received the B.S. degree from Hohai University, Nanjing, China, in 2023. He is currently pursuing the M.S. degree in pattern recognition and intelligent systems with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China.

His research interests include computer vision, neural rendering, and 3-D reconstruction.



Junyuan Liu received the B.S. degree from Dalian Maritime University, Dalian, China, in 2021. He is currently pursuing the Ph.D. degree in pattern recognition and intelligent systems with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China.

His research interests include 3-D reconstruction and image matching.



Xiaoran Yin received the B.S. degree from Tsinghua University, Beijing, China, in 2016, and the Ph.D. degree from the Institute of Metal Research, Chinese Academy of Sciences, Shenyang, China, in 2022.

She is currently an Associate Research Fellow with Shenyang Institute of Automation, Chinese Academy of Sciences. Her research interests include high-resolution reconstruction of CT scan images, 3-D reconstruction of internal defects in materials, and detection and identification of internal microdefects in materials based on CT scanning.



Zi Yang received the B.S., M.S., and Ph.D. degrees from Xidian University, Xi'an, China, in 2013, 2016, and 2021, respectively.

She is an Associate Research Fellow with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China. Her main research focuses on high-precision analysis of industrial CT scan data, 2-D and 3-D visualization technology, and detection and identification of internal microdefects in materials based on CT scanning.



Yunfeng Luo received the M.S. degree from Changchun University of Science and Technology, Changchun, China, in 2021.

He is an Engineer with Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China. His research interests include optical design, image processing, and intelligent remote sensing.



Daijun Zhang received the Ph.D. degree from Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, in 2017.

He is currently a Professor with Shenyang Institute of Automation, Chinese Academy of Sciences. His research interests include optoelectronic information systems and optical functional film.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:16:37 UTC from IEEE Xplore. Restrictions apply.